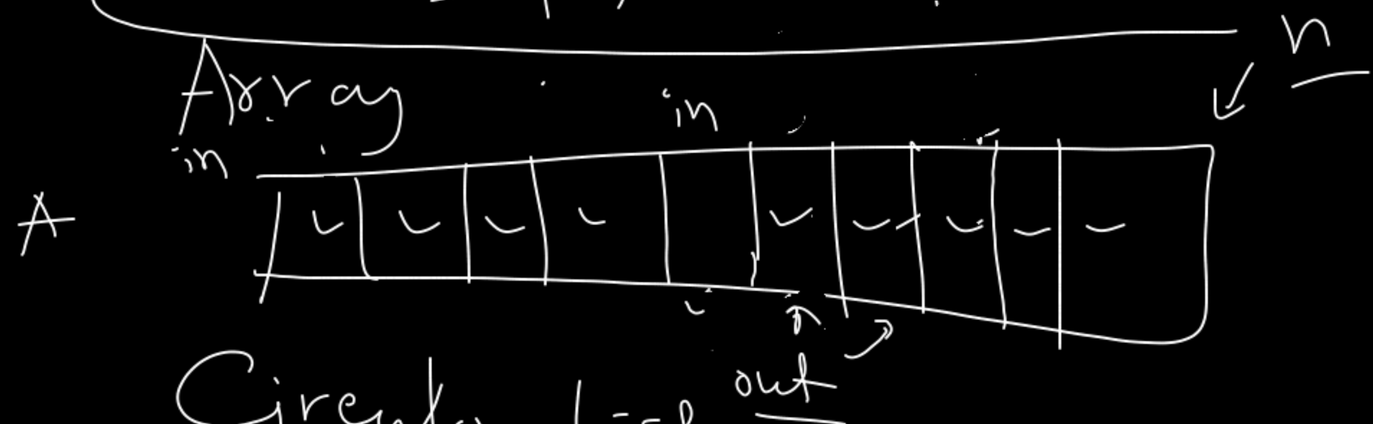
Sol^n → Fill only n-1 items.

Full → in + 1 == out

Empty → in == out

Array



Circular list

Full ⇒ in == out

Empty → in == out

Produced → A[in] = produced item

in = in + 1

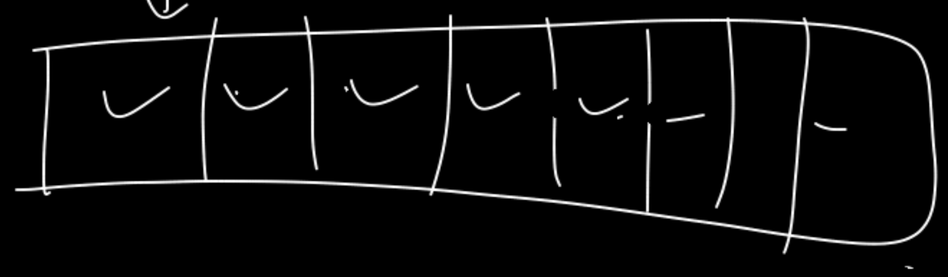
Consumption

Consumed_element = A[out]

out = out + 1

counter , = -1

-1



count =

stack

counter = counter + 1

A[counter] = new_item

consumed_item = A[counter]

counter = counter - 1

Process: Producer

Arr[n]

```
while ( True ) {
    while ( count == n ) ; // do nothing
        new_item = produced item

        Arr [count] = new item

        count = count + 1
}
```

indicate empty

Consumer

```
while (True) { while ( count == 0 ) ;
    consumed item = Arr[count]

    count = count - 1

}
```

$c = 4$. Count = Count + 1

① load $R_1 \leftarrow$ count
4

⑤ ADD $R_1 = R_1 + 1$
5

⑥ Store Count = R_1
5

Count--;

② $R_2 \leftarrow$ count
4

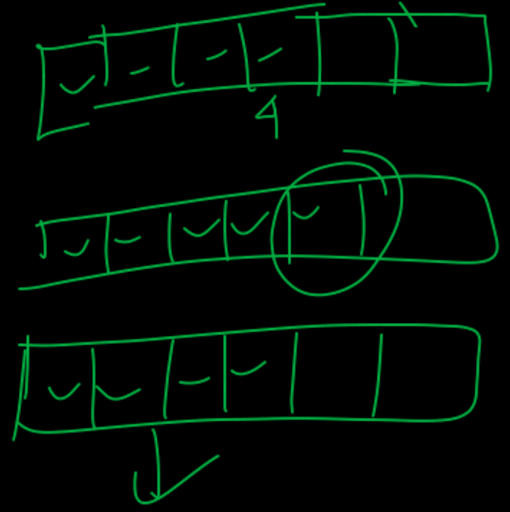
③ $R_2 = R_2 - 1$
3

④ Count = R_2
3

Consumer

Count = 4

①



Thread → Sharing $\boxed{\text{Count}}$ variable

producer

global variable

i = 0 turn = 0;

```
while (true){

    while (turn != 0);
        count = count + 1
    turn =
}
```

i = 1

```
while (True){

    while (turn != 1);
        count = count - 1
    turn = 1
}
```